Computer Engineering B.S. Degree
2025-2026 Curriculum Chart

Math Courses

- **MATH 19A** Calculus I
- **AM 10** Engr. Math Methods I *or* **MATH 21** Linear Algebra
- **MATH 19B** Calculus II
- **AM 20** Engr. Math Methods ll *or* **MATH 24** Ordinary Differential Equations
- **CSE 16** Discrete Math
- **AM 30** Multivariate Calculus for Engineers *or* **MATH 23A** Vector Calculus
- **ECE 103/L** Signals & Systems
- **CSE 107** Probability & Statistics *or* **STAT 131** Introduction to Probability Theory

Science Courses

- **PHYS 5A/L** Mechanics
- **PHYS 5C/N** Electricity & Magnetism
- **PHYS 5B/M** Waves & Optics *or* **ECE 9** Statics and Mechanics of Materials

Major Qualification Policy

Complete 36 units of the highlighted courses by the end of 5th quarter with at least a 2.8 GPA and no more than seven credits of non-passing grades in those courses. You do not need to complete all highlighted classes for major declaration.

DC Requirement

- **CSE 185E** Technical Writing *or* **CSE 195**** Senior Thesis

Core Courses

- **CSE 20** Beginning Programming in Python
- **CSE 30** Programming Abstractions: Python
- **CSE 12** Computer Systems & Assembly Lang.
- **CSE 13S*** Computer Systems & C Programming *or* **ECE 13** Computer Systems & C Programming
- **CSE 100/L** Logic Design
- **CSE 101** Intro to Data Structures and Algorithms
- **CSE 120** Computer Architecture
- **CSE 121** Embedded System Design
- **ECE 101/L** Electronic Circuits

*CSE 13s is strongly recommended, with exception to those pursuing a Digital Hardware Concentration

Concentrations (choose one)

System Programming

- **CSE 130**
- **CSE 150**
- **CSE 111** or **CSE 134**
- *One of the following:*
  - •**CSE 113**
  - •**CSE 156/L**
  - •**CSE 110A**
- **Elective** Can be chosen from the Computer Engineering Elective list on the UA website.

Computer Systems

- **CSE 130**
- **CSE 111** or **CSE 134**
- **CSE 125** *CSE 225 can be substituted with department approval.* *or* **CSE 122** *CSE 222A can be substituted with department approval*
- **Elective** Can be chosen from the Computer Engineering Elective list on the UA website.

Networks

- **CSE 130**
- **CSE 150**
- **CSE 156/L**
- **Elective** Can be chosen from the Computer Engineering Elective list on the UA website.

Digital Hardware

- **CSE 125** CSE 225 can be substituted with department approval.
- **ECE 171/L** or **CSE 122** CSE 222A may be substituted for CSE 122 with department approval
- *One of the following:*
  - **CSE 122** (if not satisfied above)
  - **CSE 220**
  - **CSE 228A**
  - **ECE 171/L** (if not satisfied above)
  - **ECE 173** (Prerequisite: ECE 174)
- **Elective** *Can be chosen from the Computer Engineering Elective list or the approved Digital Hardware Grad- Level Course List*

Elective course options can be found on the Undergrad Advising website at: https://undergrad.engineering.ucsc.edu/curriculum-charts/

Capstone (choose one option)

- **CSE 123A, 123B** Eng. Design, I & II. Prerequisites: CSE 121 and previous or concurrent enrollment in CSE 185E
- **CSE 127A, 127B** Chip Design, I & II. Prerequisites: CSE 125 or 225, and previous or concurrent enrollment in CSE 185E, and previous or concurrent enrollment in CSE 122 or 222A.
- **ECE 118 *** Intro to Mechatronics. Prerequisites: ECE 101/L, CSE 100/L, and ECE 13
- **CSE 195**** Senior Thesis. *Satisfies the DC requirement.* Special approval needed, see capstone note below

Please see catalog for full list of approved capstones. Capstones not listed on this curriculum chart have additional enrollment restrictions or are not consistently offered.

**ECE 118 can NOT be used as elective and capstone. **CSE 195 also requires the submission of an approved senior thesis.*

Computer Engineering B.S. Degree
2025-2026 Curriculum Chart

Fall ________	Winter ________	Spring ________	Summer ________

Fall ________	Winter ________	Spring ________	Summer ________

Fall ________	Winter ________	Spring ________	Summer ________

Fall ________	Winter ________	Spring ________	Summer ________

Upper Division Electives

Please refer to the Undergraduate Advising website for the list of approved electives:
https://undergrad.engineering.ucsc.edu/curriculum-charts/

The prerequisites listed on this curriculum chart are accurate as of July 22, 2025 according to UCSC's general catalog. Prerequisites listed on this chart are subject to change and students should refer to the catalog for the most up to date requirements.